

Behnam Sadeghi, PhD, FAIG, FAusIMM, FAAG, RPGeo, Fulbrighter

CSIRO, The Australian Resources Research Centre (ARRC), Australia

University of New South Wales (UNSW), Dept. of Biological, Earth and Environmental Sciences (BEES), Australia

Cell: +61 406 521 225 | behnam.sadeghi@csiro.au | behnam.sadeghi@fulbrightmail.org

<https://sites.google.com/carnegiescience.edu/dr-behnam-sadeghi/bio>

<https://people.csiro.au/S/B/behnam-sadeghi>



EDUCATION

Degree	Institution	Outcome	Years
Ph.D. of Geosciences <i>Thesis: Quantification of Uncertainty in Geochemical Anomalies in Mineral Exploration</i>	UNSW, Australia	8 journal articles 17 conference abstracts, 1 book	2018-2020
Master of Mining Engineering-Exploration <i>Thesis: Application of Concentration-Number (C-N) Fractal Method to Outline Mineralized Zones in the Zaghia Iron Ore Deposit, Bafq, Anomaly 2C</i>	IAU, Iran	5 journal articles 5 conference abstracts 1 book	2009-2012
Bachelor of Mining Engineering-Exploration <i>Thesis: Lead & Zinc Polymetallic Deposits and their Exploration Methods: Rokn Abad Lead & Zinc Deposit, Ardabil, Iran</i>	UoK, Iran	2 conference abstracts	2004-2009

SELECTED PROFESSIONAL EXPERIENCE

Fulbright Research Scholar	Carnegie Institution for Science, Earth and Planets Laboratory, Washington, D.C., USA Stanford University, School of Earth, Energy & Environmental Sciences, California, USA	2023-2024
Research Fellow	CSIRO (ARRC), Mineral Resources: Critical Minerals Exploration ◦ Leading “Geochemical & Mineralogical studies of Rubidium/Lithium distribution in pegmatitic minerals at Mt. Edon, WA”-\$110k contributions - a collaboration with Everest Metals Corporation (ASX: EMC). ◦ MinEx CRC project “Integrating Petrophysical and Mineralogical Data for Conceptual Geophysical Modelling in Orebody Characterisation, Cobar (NSW)” ◦ TIMA Texture project “Textually informed classification of igneous rocks using automated mineralogical maps”. ◦ NW Victoria Geochemistry project “Baseline geochemistry: groundwater, soil, salt lake material, and vegetation in west and northwest Victoria” – a collaboration with the Geological Survey of Victoria. ◦ Australian Scalable Drone Cloud (ASDC) monitoring project.	2023-Present

Visiting Research Fellow	University of New South Wales	2022-Present
Honorary Research Fellow	EarthByte Group, USYD	2022-2023
Principal Consultant	Independent	2022-Present
Senior Geologist & Consultant	Perpetual Resources Ltd (ASX: PEC) (WA and NSW, Australia)	2022
Senior Geologist & Data Analyst	Impact Minerals Ltd (ASX: IPT) (WA, Australia)	2021-2022
Researcher/Technical Software Supporter	EarthByte Group, USYD	2020-2021
Senior Geologist & Data Analyst/Consultant	Satrap Resources (WA)	2020-2021
Senior Geologist	Hampa Behineh Consultant (Iran)	2014-2020
Researcher	James Cook University: EGRU	2015-2017
Senior Geologist	Samaneh Kansar Zamin (Iran)	2014
Junior Geologist	Rahpira (Iran)	2011-2014
Junior Geologist	Hampa Behineh Consultant (Iran)	2007-2011

SELECTED TEACHING & MENTORING EXPERIENCE

IAMG Short Course	<i>“Applied Geochemistry and Analytics”</i>	Aug 2023
GEOL 5806 - Modular Graduate Course in Exploration Geochemistry	<i>“VMS and orogenic gold mineralization and exploration”</i> 11-day workshop, Harquail School of Earth Sciences’ Mineral Exploration Research Centre (MERC) at Laurentian University, Canada	Dec 2022
Co-advisor	Geology-machine learning projects with UNSW (Dr. R. Chandra) and interns	2021-Present
Researcher/Technical Software Supporter/ GPlates software tutorials provider	EarthByte Group, USYD	2020-2021
Mentor of academics from various fields	<i>“What is fractal geometry and what is its difference with Euclidean geometry?”</i> , UNSW	2019
Teaching/Research Assistant	IAU Mining Engineering and Geosciences Departments: Intro Geology, ArcGIS, RockWorks, Surfer, SPSS, ioGAS, Computational Geoscience.	<i>Virtual:</i> 2015-2020 <i>In-Person:</i> 2011-2015

Course Designer & Lecturer	“ <i>Fundamentals of scientific writing</i> ” Tehran Institute of Technology	2014-2015
Volunteer Instructor for Adult Literacy	Provided education to elderly individuals with limited reading and writing skills, “Khaneye Farhang” (In English: Culture House)	~1997

FUNDING & AWARDS

- Fulbright Future Scholarship (funded by The Kinghorn Foundation): USD 29,325.00 (2023-2024)
- Fulbright travel support and invitation for participating in the Fall 2023 Cleveland Fulbright Scholar Enrichment Seminar, Cleveland, Ohio, USA (2023)
- Travel support to attend AGU 2023 (San Francisco, USA), Carnegie Institution for Science: USD 3,500 (2023)
- Travel support to attend GSA 2023 (Pittsburgh, USA), Carnegie Institution for Science: USD 2,500 (2023)
- Research Fellowship, Carnegie Institution for Science: USD 16,000.00 (2023-2024)
- Andrei Borisovich Vistelius Award, International Association for Mathematical Geosciences (IAMG) (2023)
- The Australasian Institute of Mining and Metallurgy (AusIMM) Fellowship (2023)
- Association of Applied Geochemists (AAG) Fellowship (2022)
- Australian Institute of Geoscientists (AIG) Fellowship (2022)
- The EGU General Assembly 2022 Roland Schlich Travel Support (2021)
- Mineral Exploration Research Centre (MERC) Postdoctoral Fellowship to Tenure-Track (Declined – COVID-19) (2020)
- Top 3 in the "Esri Young Scholars Award" (2020)
- Top 4 in the "Esri Young Scholars Award" (2019)
- The first recipient of the 2020 "Founders Scholarship" award of the International Association for Mathematical Geosciences (IAMG) - US\$4000 (2019)
- GeoHost support grant for participating in the 36th IGC, 2-8 March 2020, Delhi, India (2019)
- UNSW BEES HDR Conference Support - \$1000 (2019)
- UNSW PRSS conference travel top-up - \$2600 (2019)
- PANGEA Student Grant- \$800 (2019)
- Australian RTP PhD Scholarship - \$27,082.00 per annum (2018-2020)
- UNSW Relocation allowance - \$1000 (2018)
- Goldschmidt 2016 Student Volunteer Role - \$500 (2018)
- Australian Professional Mining Engineer, Skill Level 1 - Engineers Australia (2016)
- International Association for Mathematical Geosciences (IAMG) Student Travel Grants - US\$1000 (2015)
- James Cook University Postgraduate Research Scholarship (PRS) - \$29,000 per year (2015-2017)
- Dr. William S. Lewis Doctoral Fellowship (Canada) - \$25,000 per year for four years (2015)
- Best Master's Thesis, Research Week, IAU (2013)
- IAU Publishing Grant - \$1500 (2012-2015)
- First rank MSc Student, Mining Engineering-Exploration, IAU (of 36 students) (2012)
- Fourth rank National Entrance Exam, MSc Mining Engineering, IAU (2008)
- University of Kurdistan Full scholarship for 4 years of BSc degree in Mining Engineering-Exploration (2004)

SELECTED CERTIFICATIONS - CERTIFICATES AVAILABLE UPON REQUEST

Certificate	Institution/Instructors	Years
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Learn to Lead with UNSW 2025: Strategic Thinking for Impactful Leadership	UNSW	Nov 2025
Standard Mental Health First Aid	CSIRO/Australian Red Cross	April 2025
Provide First Aid	CSIRO/Australian Red Cross	March 2025
Professional Certificate in JORC Code Reporting	AusIMM	Jan 2025
Using GenAI Effectively at Work: Generative AI Skills for Early Career Researchers and STEM Professionals	CSIRO	Nov 2024
Python for Geoscientists	Curtin University	Nov 2024
Regolith Geology and Geochemistry training course	CSIRO	Nov 2024
Field Medication Kit Training	CSIRO	Sep 2024
Competitive CVs for Careers and Opportunities - Virtual	CSIRO	Sep 2024
Unsupervised Learning in Python	DataCamp	Feb 2024
Supervised Learning with scikit-learn	DataCamp	Feb 2024
Data manipulation with pandas	DataCamp	Feb 2024
Fulbright Visiting Scholar Workshop on Scientific Manuscript Publishing & Peer Reviews	Fulbright	Mar 2024
PMP Certification Exam Prep Course	Udemy	Dec 2024
Introduction to Professional Grant Development Workshop	Fulbright	Nov 2023
Registered Professional Geoscientist (RPGeo 10295) in the specific field of Information Geoscience	AIG	Aug 2023
Successful Project Management at CSIRO	CSIRO	Jul 2023
Project Leader Induction: Module 3- Communication & Leadership	CSIRO	Jun 2023
Project Leader Induction: Module 2- Managing a Project Lifecycle in Dynamics	CSIRO	Jun 2023
Project Leader Induction: Module 1- Introduction to Project Management	CSIRO	May 2023
Writing Winning Scientific Proposals	CSIRO	May 2023
Publishing with Impact	CSIRO	May 2023
Autism in the Workplace - AMAZE	CSIRO	May 2023
Practical Geocommunication	Geologize Ltd	Oct 2022
Truth Telling: Racism and Reconciliation (to support EDI)	Fulbright Australia	Feb 2022
Data8.1x: Data Science: Computational Thinking with Python	edX-University of California, Berkeley	Jul 2021
Spatial Data Science: The New Frontier in Analytics (Esri ArcGIS Pro)	Esri	Dec 2020
Founders Start! (UNSW Founders Program)	UNSW	Aug 2019

ON Program (Link academia & industry)	CSIRO	Aug 2019
Beginning to Teach (BTT) Program for ECRs and HDRs	UNSW	Jun 2019
Provide First Aid (four courses)	UNSW/Allens Training	Mar 2019
Focus on Peer Review	Springer Nature- Nature Masterclasses	May 2019
Mental Health First Aid	Mental Health First Aid Australia (MHFA)	Feb 2019
Machine Learning	Coursera-Stanford University	Jan 2019
The Google Fundamentals of Digital Marketing	Google Digital Garage	Dec 2018
Writing Clear Science (Dr. Marina Hurley)	UNSW	Dec 2018
Mining engineering - exploration and performance, Grade 3	Iranian Mining Eng. Organization (IMEO)	Mar 2017
Introduction to MATLAB	Tehran Institute of Technology	Jul 2014
Remote sensing using satellite images (ENVI software)	Samaneh Negar Zamin Co.	Nov 2012
Datamine Software	Samaneh Negar Zamin Co.	Oct 2012
Writing Scientific Papers	Samaneh Negar Zamin Co.	Sep 2012
Geochemistry of stable isotopes and its application in geology, with a special attitude to economic geology Studies	Shahid Beheshti University	Jan 2012
An introduction to Geolog software	Iranian Society of Petroleum Geology (ISPG)	Dec 2011
Geochronology and Thermochronology with Applications to Intracontinental Extension and Formation of Metamorphic Core Complexes	Dr. Jamshid Hasanzadeh (USA), Geological Survey of Iran	Nov 2011
Modern Techniques in Mapping of Geochemical Anomalies and Mineral Prospectivity	Prof. Emmanuel John M. Carranza (Australia), Shahrood University of Technology	Nov 2011
Advanced ArcGIS Software	Samaneh Negar Zamin Co.	Nov 2011
Basic ArcGIS Software	Samaneh Negar Zamin Co.	Oct 2011
Geomathematical Mineral Resource Modeling and Prediction	Prof. Frits Agterberg (Canada), University of Tehran	Oct 2011
Geomathematical Mineral Resources Modeling Based on copper deposits	Prof. Frits Agterberg (Canada), First World Copper Congress	Oct 2011
Porphyry Copper	Prof. Bernd Lehman (Germany), First World Copper Congress	Oct 2011
Radiolarians and its importance in Biostratigraphy Studies	Dr. Mohsen Kariminia (USA), Geological Survey of Iran, Iran	Oct 2011
RockWorks Software	Samaneh Negar Zamin Co.	Sep 2011
Stratigraphy, Structure Geology & no Temperature Thermochronology across the Makran Accretionary Wedge in Iran	Prof. John Pierre Burg (Switzerland), Geological Survey of Iran	Jan 2011
Earth-Surface Processes	Prof. Allan Ross Chivas (Australia), Geological Survey of Iran	Sep 2010

Quantitative Active Tectonic Studies	Profs. Richard Walker, Morteza Talebian, & Morteza Fattahi (UK), Geological Survey of Iran	Jun 2010
Giant Metallic Deposits Including Gold, VMS and SEDEX	Prof. Peter Laznicka (Australia), Geological Survey of Iran	May 2010
Granite-Related Ore Deposits (Workshop and Field Trip)	Prof. Bernd Lehman (Germany), Geological Survey of Iran	Mar 2010
Mineral Resources: Geology and Geochemistry of Ore Deposits (Workshop and Field Trip)	Prof. Bernd Lehman (Germany), Geological Survey of Iran	Feb 2010
Initial and final mineral exploration planning	Industries and Mines Organization of Kurdistan	Sep 2009
GPS	Industries and Mines Organization of Kurdistan	Aug 2009
Retraining technical authorities in the mines	Industries and Mines Organization of Kurdistan	Aug 2009

INVITED LECTURES AND EVENTS

- Invited [keynote lecture](#), *Shaping the Future of Exploration: Decision-Making Under Geological Uncertainty*, [MinProXT](#) 2025 webinar (Geological Survey of Finland and Canada, 2025).
- Invited distinguished referee to provide the Royal Irish Academy with an objective assessment of the merits of one of the candidates under consideration for membership (Invited by Professor Daniel Carey on behalf of the Academy and its Advisory Committee), Ireland (2024).
- Invited Mining/Geology Professional with a team of entrepreneurs as part of the National science Foundation (NSF)'s National I-CORPS program, Stanford University, USA (2024)
- Invited lecture, *Unlocking the Hidden Codes in Mineral Exploration*, Mineral-X, Doerr School of Sustainability, Stanford University, USA (To be announced!)
- VIP invited attendee, [Mineral-X Symposium 2024](#): *Critical Minerals: Efficient Exploration and Responsible Mining*, Paul Brest Hall, Doerr School of Sustainability, Stanford University, USA (2024)
- Invited lecture, *Mastering Uncertainty in Mineral Exploration for the Energy Transition*, University of Connecticut (UConn), Department of Earth Sciences, USA (2024)
- Invited Fulbright Scholar, Next Gen Leaders' Reception, Embassy of Australia in Washington, D.C., USA (2024)
- Invited Fulbright Scholar, Fulbright Gala by Fulbright NCAC at the Embassy of Austria in Washington, D.C., USA (2024)
- Invited Fulbright Scholar, *Health disparities and the social determinants of health through a lens of education and community engagement*, the Fall 2023 Cleveland Fulbright Scholar Enrichment Seminar, Cleveland, Ohio, USA (2023)
- [Keynote lecture](#), *Decision-making for follow-up mineral exploration based on spatial uncertainty of geochemical data*, [IAMG 2023](#)'s Andrei Borisovich Vistelius Research Award, Trondheim, Norway (2023)
- Invited lecture, *Uncertainty Quantification and Modeling*, [MineTED 2023](#), online (2023)
- Invited lecture, *How to use a decision-making process to target geochemical anomalies based on frequency and Bayesian frameworks?*, [Target 2023 Conference](#), Perth, Australia (2023)
- Invited Fulbright Scholar, Fulbright Gala by Fulbright Australia at the Australian Parliament House in Canberra, Australia (2022)
- Invited lecture, *New insights to support Australia's Exploring for Future strategies*, Geoscience Australia, online (2022)
- Invited lecture, *How new and novel datasets and techniques could be used to improve the robustness of mineral potential mapping in NSW?*, New South Wales Government, online (2022)

- Invited lecture, *Fractal and multifractal modeling and the relevant uncertainty analysis*, Mineral Exploration Research Centre (MERC), Laurentian University, online (2022)
- Invited lecture, *Mineral exploration VMS and Au case studies*, Mineral Exploration Research Centre (MERC), Laurentian University, online (2022)
- Invited lecture, *Introduction, ideas, plans, and projects in data science applied to geosciences*, the Sustainable Minerals Institute (SMI), University of Queensland (2022)
- [Keynote lecture](#), *All models are wrong, but what “some” are useful?*, [IAMG Founders Scholarship Award winner](#) (2021)
- Invited lecture, *My insights about research and teaching in Geospatial Science*, Cardiff University, online (2021)
- Invited lecture, *Fractals and Multifractals in Exploration Geochemistry and Anomaly Detection*, EarthByte Group, University of Sydney (2020).

PUBLICATIONS

PEER-REVIEWED PAPERS:

1. Acosta-Gongora, P., Andersson, M., Bjerkgård, T., Morris, W.M., Kurz, T., Lee, M., Dumais, M.A., Nasuti, A., van Boeckel, M., Jakob, J., Miranda, A.C.R., Crilly, A., Wang, Y., **Sadeghi, B.** *, 2025 ‘, *Journal of Geochemical Exploration*, Vol. 280, 107911 (<https://doi.org/10.1016/j.gexplo.2025.107911>).
2. Bajadi, A., Ouchchen, M., El Azzab, D., Achkouch, L., Dadi, B., Abia, E.H., **Sadeghi, B.** *, 2025 ‘Targeting potential cobalt deposits through integration of helicopter-borne magnetic and stream sediment data: Insights from the Bou Azzer Inlier, Central Anti-Atlas, Morocco’, *Ore Geology Reviews, Special Issue: Critical metal deposits within magmatic-hydrothermal systems*, vol. 185, 106807 (<https://doi.org/10.1016/j.oregeorev.2025.106807>).
3. Heidari, S.M., Afzal, P., **Sadeghi, B.** *, 2025 ‘Corrected the mathematical formula for Euler deconvolution, Revised the formula for the fuzzy logic model, Added the reference to the geological map’, *Journal of Geochemical Exploration* (<https://doi.org/10.1016/j.gexplo.2025.107854>).
4. **Sadeghi, B.** *, 2025 ‘Clustering in Geo-Data Science: Navigating Uncertainty to Select the Most Reliable Method’, *Ore Geology Reviews*, 181C, 106591 (<https://doi.org/10.1016/j.oregeorev.2025.106591>).
5. Afzal, P., Heidari, B., Alidaryan, M., **Sadeghi, B.**, Daneshvar Saein, L., 2025 ‘Enhancing core recovery and RQD analysis in porphyry Cu deposits through fractal modeling, case study: Sungun mine (NW Iran)’, *International Journal of Mining and Geo-Engineering*, vol. 59 (1), p.p. 69–74 (<https://doi.org/10.22059/ijmge.2025.383290.595200>).
6. Achkouch, L., Attou, A., Mezougane, H., Ouchchen, M., El Cheikh, Y., Mamouch, Y., Bajadi, A., Dadi, B., Ahmed, R., Sadeghi, B., 2025 ‘Integration of Multi-Method ASTER data Analysis and Geometric Average Modeling for Hydrothermal Alteration Mapping and Mineral Prospectivity Assessment of Copper Deposits, Anti-Atlas, Morocco’, *Remote Sensing Applications: Society and Environment*, 37, 101467 (<https://doi.org/10.1016/j.rsase.2025.101467>).
7. Dadi, B., Ouchchen, M., Faik, F., Boutaleb, S., El Azzab, D., Mamouch, Y., Achkouch, L., Bajadi, A., Abia, E.H., **Sadeghi, B.** *, 2024 ‘Hydrothermal Fluid Pathways and Mineralization Potential in the High Atlas Massif (Morocco) Using Fuzzy Logic and Multifractal Modeling’, *Ore Geology Reviews*, 175, 106401 (<https://doi.org/10.1016/j.oregeorev.2024.106401>).
8. Eskandari, A., Amini, S., **Sadeghi, B.** *, 2024 ‘The dynamic of magmatic system and volcano hazard implications of the Damavand volcano (N. Iran) inferred from the textural data’, *Journal of Volcanology and Geothermal Research*, 456C, 108220 (<https://doi.org/10.1016/j.jvolgeores.2024.108220>).
9. **Sadeghi, B.** *, Eleish, A.M., Morrison, S.M., Klump, J., 2024 ‘Inverse radius weighting and its python package “IRWPpy”: A new topography-informed interpolation to enhance geological interpretations’, *Ore Geology Reviews*, 172, 106206 (<https://doi.org/10.1016/j.oregeorev.2024.106206>).

10. Eskandari, A., **Sadeghi, B.** *, 2024 ‘Deciphering Igneous Rock Crystals: Unveiling Multifractal Patterns in Crystal Size Dynamics’, special issue *"Texture and Microstructural Analysis of Crystalline Solids, Volume II"*, *Minerals*, 14(7), 660 (<https://doi.org/10.3390/min14070660>).
11. **Sadeghi, B.** *, Jowitt, S., Porwal, A., Lucci, F., 2024 ‘Preface: “Critical Minerals: Current Perspectives”’, *Geochemistry, Special Issue on “Critical Minerals: Current Perspectives”*, 84(2), 126131 (<https://doi.org/10.1016/j.chemer.2024.126131>).
12. Heidari, S.M., Afzal, P., **Sadeghi, B.** *, 2024 ‘Molybdenum and gold distribution variances within Iranian copper porphyry deposits’, *Journal of Geochemical Exploration, Special Issue on “Multi-element Geochemical Mapping in South and Central Asia at National Scale”*, 261, 107471 (<https://doi.org/10.1016/j.gexplo.2024.107471>).
13. Esmailoghli, S., Lima, A., **Sadeghi, B.** *, 2024 ‘Lithium exploration targeting through robust variable selection and deep anomaly detection: An integrated application of sparse principal component analysis and stacked autoencoders’, *Geochemistry, Special Issue on “AI-driven mineral exploration”*, 126111 (<https://doi.org/10.1016/j.chemer.2024.126111>).
14. Bajadi, A., El Azzab, D., Ouchchen, M., Mamouch, Y., Achkouch, L., Dadi, B., El Moutaouakkil, N., **Sadeghi, B.** *, 2024 ‘Assessment of potential Cobalt zones by modeling radiometric and ASTER data using the AHP method: a case of the Bou Azzer El Grara buttonhole, Central Anti-Atlas, Morocco’, *Geochemistry, Special Issue on “Critical Minerals: Current Perspectives”*, 84(2), 126105 (<https://doi.org/10.1016/j.chemer.2024.126105>).
15. **Sadeghi, B.** *, Molayemat, H., Pawlowsky-Glahn, V., 2024 ‘How to choose a proper representation of compositional data for mineral exploration?’, *Journal of Geochemical Exploration, Special Issue on “Compositional Data Analysis 40 years after the paper of 1982 by John Aitchison: state of the art, perspectives and developments”*, 259, 107425 (<https://doi.org/10.1016/j.gexplo.2024.107425>).
16. Akbarpour, A., Ghalamghash, J., Azizi, H., **Sadeghi, B.** *, 2024 ‘Mineralogy, geochemistry and mineralization of Maimoonabad iron ore deposit, northwest of Sanandaj-Sirjan zone, Iran’, *Journal of Geochemical Exploration*, 259, 107427 (<https://doi.org/10.1016/j.gexplo.2024.107427>).
17. Akbarpour, A., Vanaei, M., **Sadeghi, B.** *, 2023 ‘Geology, mineralogy, and alteration of the Cu-Pb veins, and their significance in mineral exploration: a case study in the Khajeh-Nezam area, SE Iran’, *Geochemistry, Special Issue on “Critical Minerals: Current Perspectives”* (<https://doi.org/10.1016/j.chemer.2023.126041>).
18. **Sadeghi, B.** *, Cohen, D.R., 2023 ‘Decision-making within geochemical exploration data based on spatial uncertainty- A new insight and a futuristic review’, *Ore Geology Reviews*, Volume 161, October 2023, 105660 (<https://doi.org/10.1016/j.oregeorev.2023.105660>).
19. Afzal, P., Gholami, H., Madani, N., Yasrebi, A.B., **Sadeghi, B.** *, 2023 ‘Mineral resource classification using geostatistical and fractal simulation in the Masjed Daghi Cu-Mo porphyry deposit, NW Iran’, *Minerals, Special Issue on “Advanced Data Analytics in Earth and Environmental Sciences”*, 13, 370. (<https://doi.org/10.3390/min13030370>).
20. Heidari, M., Afzal, P., **Sadeghi, B.** *, 2023 ‘Miocene tectono-magmatic events and gold mineralization in the Takab-Delijan belt, Iran’, *Geochemistry*, 83(1), 125944 (<https://doi.org/10.1016/j.chemer.2022.125944>).
21. Saein, L.D., Afzal, P., Shahbazi, S., **Sadeghi, B.** *, 2022 ‘Application of an improved zonality index model integrated with multivariate fractal analysis: epithermal gold deposits’, *Geopersia*, 12(2), pp. 379–394 ([DOI: 10.22059/GEOPE.2022.339864.648652](https://doi.org/10.22059/GEOPE.2022.339864.648652)).
22. **Sadeghi, B.** *, 2022 ‘Chatterjee Correlation Coefficient: A robust alternative for classic correlation methods in geochemical studies - (including “TripleCpy” Python package)’, *Ore Geology Reviews*, vol. 146, 104954 (<https://doi.org/10.1016/j.oregeorev.2022.104954>).
23. Farhadi, S., Afzal, P., Boveiri Konari, M., Daneshvar Saein, L., **Sadeghi, B.** *, 2022 ‘Combination of Machine Learning Algorithms with Concentration-area Fractal Method for Soil Geochemical Anomaly Detection in

Sediment-Hosted Irankuh Pb-Zn Deposit, Central Iran', *Minerals: section of "Mineral Exploration Methods and Applications"*, vol. 12(6), 689 (<https://doi.org/10.3390/min12060689>).

24. Yilmaz, H., **Sadeghi, B.**, Cohen, D.R., 2022 'The efficiency of fractal techniques in geochemical anomaly delineation within BLEG and <180 µm stream sediments in Western Turkey', *Journal of Geochemical Exploration* (<https://doi.org/10.1016/j.gexplo.2022.106957>).
25. Wang, H., Yuan, Z., Cheng, Q., Zhang, S., **Sadeghi, B.** *, 2022 'Geochemical anomaly definition using stream sediments landscape modeling', *Ore Geology Reviews* (<https://doi.org/10.1016/j.oregeorev.2022.104715>).
26. Daneshvar Saein, L., Afzal, P., Shahbazi, S., **Sadeghi, B.** 2022 'Application of an improved zonality index model integrated with multivariate fractal analysis: epithermal gold deposits', *Geopersia* (In press).
27. Heidari, S.M., Afzal, P., Ghaderi, M., **Sadeghi, B.** *, 2021 'Detection of mineralization stages using zonality and multifractal modeling based on geological and geochemical data in the Au-(Cu) intrusion-related Gouzal-Bolagh deposit, NW Iran', *Ore Geology Reviews* (<https://doi.org/10.1016/j.oregeorev.2021.104561>).
28. **Sadeghi, B.** *, Zhao, H., Holzheid, A., 2021 'Preface: Mineral exploration: a journey from fieldwork, to laboratory work, computational modelling and mineral processing', *Geochemistry, Special Issue on "Mineral exploration: a journey from fieldwork, to laboratory work, computational modelling and mineral processing"*, vol. 81, Issue 4 (<https://doi.org/10.1016/j.chemer.2021.125825>).
29. **Sadeghi, B.** *, 2021 'Simulated-multifractal models: a futuristic review of multifractal modeling in geochemical anomaly classification', *Ore Geology Reviews*, 139 (Part B) (<https://doi.org/10.1016/j.oregeorev.2021.104511>).
30. **Sadeghi, B.** *, Cohen, D.R., 2021 'Concentration-distance from centroids (C-DC) multifractal modeling: A novel approach to characterizing geochemical patterns based on sample distance from mineralization', *Ore Geology Reviews* (<https://doi.org/10.1016/j.oregeorev.2021.104302>).
31. **Sadeghi, B.** *, Cohen, D.R., 2021 'Category-based fractal modelling: A novel model to integrate the geology into the data for more effective processing and interpretation', *Journal of Geochemical Exploration* (<https://doi.org/10.1016/j.gexplo.2021.106783>).
32. Zissimos, A.M., Cohen, D.R., Christoforou, I.C., **Sadeghi, B.**, Rutherford, N.F., 2021 'Controls on soil geochemistry fractal characteristics in Lemesos (Limassol), Cyprus', *Journal of Geochemical Exploration*, vol. 220, (<https://doi.org/10.1016/j.gexplo.2020.106682>).
33. **Sadeghi, B.** *, 2021 'SPS Model: a significant algorithm to reduce the time and computer memory required in geostatistical simulations', *Iranian Journal of Earth Sciences*, vol. 13(1), pp. 21–30 (http://ijes.mshdiau.ac.ir/article_680583.html).
34. Mokhtari, Z., **Sadeghi, B.**, 2021 'Geochemical anomaly definition using multifractal modeling, validated by geological field observations: Siah Jangal area, SE Iran', *Geochemistry, Special Issue on "Mineral exploration: a journey from fieldwork, to laboratory work, computational modelling and mineral processing"*, vol. 81, Issue 4 (<https://doi.org/10.1016/j.chemer.2021.125774>).
35. **Sadeghi, B.** *, 2021 'Concentration-concentration fractal modelling: a novel insight for correlation between variables in response to changes in the underlying controlling geological-geochemical processes', *Ore Geology Reviews* (<https://doi.org/10.1016/j.oregeorev.2020.103875>).
36. **Sadeghi, B.** *, Yilmaz, H., Pirajno, F., 2021 'Weighting of BLEG data with drainage and catchment properties to enhance Au anomalies', *Geochemistry* (<https://doi.org/10.1016/j.chemer.2020.125733>).
37. Madani, N., **Sadeghi, B.**, 2019 'Capturing Hidden Geochemical Anomalies in Scarce Data by Fractal Analysis and Stochastic Modelling'. *Natural Resources Research*, vol. 28, no. 3, pp. 833–847.
38. Khalajmasoumi, M., **Sadeghi, B.** *, Carranza, E.J.M., Sadeghi, M., 2017 'Geochemical anomaly recognition of rare earth elements using multi-fractal modelling correlated with geological features, Central Iran'. *Journal*

of Geochemical Exploration, Special issue on “Critical metals in the Middle East and North Africa – Geochemistry: Exploration and Analysis”, vol. 181, pp. 318–332.

39. Rahimzadegan, M., **Sadeghi, B.**, 2016 ‘Development of the iterative edge detection method applied on blurred satellite images: state of the art’, *Journal of Applied Remote Sensing*, vol. 10 (3) 035018 doi: 10.1117/1.JRS.10.035018.
40. **Sadeghi, B.** *, Khalajmasoumi, M., 2015 ‘A Futuristic Review for Evaluation of Geothermal Potentials Using Fuzzy Logic and Binary Index Overlay in GIS Environment’, *Renewable & Sustainable Energy Reviews*, vol. 43C, pp. 818–831.
41. **Sadeghi, B.** *, Madani, N., Carranza, E.J.M., 2015 ‘Combination of geostatistical simulation and fractal modelling for mineral resource classification’, *Journal of Geochemical Exploration*, vol. 149, pp. 59–73.
42. Nazarpour, A., **Sadeghi, B.**, Sadeghi, M., 2015 ‘Application of fractal models to characterization and evaluation of vertical distribution of geochemical data in Zarshuran gold deposit, NW Iran’, *Journal of Geochemical Exploration*, vol. 148, pp. 60–70.
43. Nazarpour, A., Rashidnejad Omran, N., Rostami Paydar, G., **Sadeghi, B.**, Matroud, F., Mehrabi Nejad, A., 2015 ‘Application of classical statistics, logratio transformation and multifractal approaches to delineate geochemical anomalies in the Zarshuran gold district, NW Iran’, *Chemie der Erde / Geochemistry*, vol. 75, pp. 117–132.
44. Rahmati, A., Afzal, P., Abrishamifar, S.A., **Sadeghi, B.**, 2015 ‘Application of Concentration-Number and Concentration-Volume fractal models to delineate mineralized zones in the Sheytoor iron deposit, Central Iran’, *Arabian Journal of Geosciences*, vol. 8, pp. 2953–2965.
45. Khalajmasoumi, M., Lotfi, M., Memar Kochebagh, A., Khakzad, A., Afzal, P., **Sadeghi, B.**, Ziazarifi, A., 2015 ‘Delineation of the radioactive elements based on the radiometric data using concentration-area fractal method in the Saghand area, Central Iran’, *Arabian Journal of Geosciences*, vol. 8, pp. 6047–6062.
46. Khalajmasoumi, M., Lotfi, M., Memar, A., Afzal, P., **Sadeghi, B.**, Khakzad, A., 2015 ‘Separation of Geochemical Anomalies of Rare Earth Elements by Concentration-Area Fractal Model in Anomaly No. 5 of Saghand (Central Iran)’, *Journal of Nuclear Science and Technology*, vol. 71, pp.113–126.
47. Rahimzadegan, M., **Sadeghi, B.**, Masoumi, M., Taghizadeh Ghalehjoghi, S., 2015 ‘Application of target detection algorithms to identification of iron oxides using ASTER images: a case study in the North of Semnan province, Iran’, *Arabian Journal of Geosciences*, vol. 8, issue 9, pp. 7321–7331.
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49. Shamseddin Meigoony, M., Afzal, P., Gholinejad, M., Yasrebi, A.B., **Sadeghi, B.**, 2014 ‘Delineation of geochemical anomalies using factor analysis and multifractal modelling based on stream sediments data in Sarajeh 1:100,000 sheet, Central Iran’, *Arabian Journal of Geosciences*, vol. 7, pp. 5333–5343.
50. Momeni, S., Shahrokhi, V., Afzal, P., **Sadeghi, B.**, Farhadinejad, T., Nikzad, M.R., 2014 ‘Delineation of the Cr mineralization based on the stream sediment data utilizing fractal modelling and factor analysis in the Khoy 1:100,000 sheet, NW Iran’, *Bulletin of the Mineral Research and Exploration*, vol. 152, pp. 143–151.
51. **Sadeghi, B.**, Khalajmasoumi, M., Afzal, P., Moarefvand, P., 2014 ‘Discrimination of Iron High Potential Zones at the Zaghia Iron Ore Deposit, Bafq, Using Index Overlay GIS Method’, *Iranian Journal of Earth Sciences*, vol. 6(2), pp. 91–98.
52. **Sadeghi, B.** *, Khalajmasoumi, M., Afzal, P., Moarefvand, P., Yasrebi, A.B., Wetherelt, A., Foster, P., Ziazarifi, A., 2013 ‘Using ETM⁺ and ASTER sensors to identify iron occurrences in the Esfordi 1:100000 mapping sheet of Central Iran’, *Journal of African Earth Sciences*, vol. 85, September 2013, pp. 103–114.

53. Afzal, P., ZiaZarifi, A., **Sadeghi, B.**, 2013 ‘Separation of geochemical anomalies using factor analysis and concentration-number (C-N) fractal modelling based on stream sediments data in Esfordi 1:100,000 sheet, Central Iran’, *Iranian Journal of Earth Sciences*, vol. 5, pp. 100–110.
54. **Sadeghi, B.** *, Moarefvand, P., Afzal, P., Yasrebi, A.B., Daneshvar Saein, L., 2012 ‘Application of fractal models to outline mineralized zones in the Zaghia iron ore deposit, Central Iran’, *Journal of Geochemical Exploration, Special Issue on “fractal/multifractal modelling of geochemical data”*, vol. 122, pp. 9–19.
55. **Sadeghi, B.** *, Moarefvand, P., Afzal, P., 2012 ‘Determination of Fe grade distribution using concentration-number fractal method in boreholes of Zaghia iron ore deposit, Bafq’, *Journal of Earth and Resources*, 4th year | 3rd issue | Autumn 2011(Series No. 12); pp. 51–60. (In Persian with English abstract).
56. **Sadeghi, B.** *, Noorollahi, Y., Radan, M.Y., Khalajmasoumi, M., 2012 ‘Evaluation of geothermal potentials using Index Overlay and Fuzzy Logic methods, Eastern Azarbayegan province 1:250,000 mapping sheet’, *Journal of Exploration Geology*, Vol.16, 5th year | 3rd issue, pp. 27–38.
57. Daneshvar Saein, L., Rasa, I., Rashidnejad Omran, N., Moarefvand, P., Afzal, P., **Sadeghi, B.**, 2012 ‘Application of number-size (N-S) fractal model to quantify of the vertical distributions of Cu and Mo in Nowchun porphyry deposit (Kerman, SE Iran)’. *Archives of Mining Sciences*, vol. 58 (2013), no 1, pp. 89–105.
58. Shamseddin Meigoony, M., Afzal, P., Gholinejad, M., **Sadeghi, B.**, 2012 ‘Processing and Comparison of Airborne Magnetic and ETM⁺ Data for Identification of Alteration Zones, Sarajeh 1:100000 Sheet, Qom’, *Journal of Exploration Geology*, 5th year | 4th issue, pp. 31–43.

CONFERENCE ABSTRACTS

ORAL PRESENTATIONS:

1. **Sadeghi, B.***, 2025 ‘Shaping the Future of Exploration: Decision-Making Under Geological Uncertainty’, *MinProXT 2025 webinar* (Geological Survey of Finland and Canada).
2. **Sadeghi, B.***, 2025 ‘Fractal Frontiers in Geosciences: Integrating Novel Multifractal Models for Enhanced Spatial Analysis’, the 23rd *International Association for Mathematical Geosciences (IAMG) conference, Zhuhai, China*.
3. **Sadeghi, B.***, Zhen Yin, David, 2025 ‘Data-Driven Innovations for Enhanced Decision-Making in Mineral Exploration’, the 23rd *International Association for Mathematical Geosciences (IAMG) conference, Zhuhai, China*.
4. Laukamp, C., **Sadeghi, B.**, 2024 ‘navigating uncertainties of remote sensing satellite image analysis in exploration geochemistry and mineralogy’, *IAGS 2024, Adelaide, Australia*.
5. **Sadeghi, B.** *, 2024 ‘Fractals and Multifractals in Geosciences: An Introduction to a New Reference Book’, *IGC 2024, South Korea* (couldn’t attend in person).
6. **Sadeghi, B.** *, Morrison, S., Eleish, A., and Klump, J., 2024 ‘Inverse Radius Weighting (IRW) interpolation model: A new interpolation considering morphology’, *EGU General Assembly 2024, Vienna, Austria*, 14–19 Apr 2024, EGU24-1348, <https://doi.org/10.5194/egusphere-egu24-1348>.
7. **Sadeghi, B.** *, 2023 ‘Decision-making for follow-up mineral exploration based on spatial uncertainty of geochemical data’, *International Association for Mathematical Geosciences (IAMG) conference, Trondheim, Norway*.
8. **Sadeghi, B.** *, 2023 ‘How to use a decision-making process to target geochemical anomalies based on frequency and Bayesian frameworks?’, *Target 2023 conference, Targeting for a New Era of Discovery*.

9. Madani, N., **Sadeghi, B.**, Maleki, M., 2023 'Evaluation of the consistency of geostatistical algorithms for resource modelling of critical elements', *WMC 2023*, Brisbane, Australia (ISBN: 978-0-646-87565-1).
10. Wang, H., **Sadeghi, B.**, 2022 'Discrimination of geochemical anomalies using geological constraints and geochemical landscape modeling', *Mineral Prospectivity and Exploration Targeting–MinProXT 2022 Webinar*, [Conference Proceedings](#), pp. 41–45.
11. **Sadeghi, B.** *, Grunsky, E., 2022 'Predictive Lithologic Mapping and Prediction of Potential Mineral Resources using a Probabilistic Approach with Multivariate Methods: a case study in Western Yilgarn, Australia', *International Association for Mathematical Geosciences (IAMG) conference*, Nancy, France.
12. **Sadeghi, B.** *, Cohen, D., Müller, D., 2022 'Improved decision-making in geochemical sampling based on both frequency and Bayesian frameworks', *EGU General Assembly 2022*, Vienna, Austria, 23–27 May 2022, EGU22-14, <https://doi.org/10.5194/egusphere-egu22-14>.
13. Farahbakhsh, E., **Sadeghi, B.**, Müller, D., Chandra, R., 2022 'An improved generative adversarial network for mapping geochemical anomalies', *16th Biennial Meeting SGA 2022*, Rotorua, NZ.
14. **Sadeghi, B.** *, 2021 'All models are wrong, but what “some” are useful?', *International Association for Mathematical Geosciences (IAMG) webinar*.
15. **Sadeghi, B.** *, Cohen, D.R., 2021 'Application of C-C fractal modelling to separate anthropogenic from geogenic patterns', *Mineral Prospectivity and Exploration Targeting–MinProXT 2021 Webinar*, [Conference Proceedings](#), pp. 27–30.
16. Farahbakhsh E., Hezarkhani A., Bahroudi A., **Sadeghi B.**, Chandra R., 2021 'Three-dimensional weights of evidence modeling of concealed ore deposits using 3DWofE software package', *Mineral Prospectivity and Exploration Targeting–MinProXT 2021 Webinar*, [Conference Proceedings](#), pp. 55–58.
17. **Sadeghi, B.** *, 2021 'Evaluation of geochemical anomaly classification models based on the relevant uncertainties and error propagation per class to select the most robust model(s) for the follow-up exploration', *EGU General Assembly 2021*, online, 19–30 Apr 2021, EGU21-1429, <https://doi.org/10.5194/egusphere-egu21-1429>, 2021.
18. **Sadeghi, B.** *, Cohen, D.R., 2020 'Simulated-fractal classification of geochemical anomalies: Sweden case study', *36th International Geological Congress (IGC)*, New Delhi, India.
19. **Sadeghi, B.** *, Cohen, D.R., 2019a 'Concentration-concentration (C-C) fractal model: a model to recognize the geochemical anomalies of target elements based on their paragenesis concentrations', *International Association for Mathematical Geosciences (IAMG) conference*, State College, Pennsylvania, USA (Conference Proceedings in Press).
20. **Sadeghi, B.** *, Cohen, D.R., 2019b 'Concentration-Distance from Centroids (C-DC) multi-fractal model: a model to recognize geochemical anomalies based on the distance of samples from the known mineralization areas', *International Association for Mathematical Geosciences (IAMG) conference*, State College, Pennsylvania, USA (Conference Proceedings in Press).
21. Sanchez, F., **Sadeghi, B.**, 2018 'Multi-fractal modelling: a significantly useful method to recognize geochemical anomalies in large-scale sampling networks', *International Association for Mathematical Geosciences (IAMG) Congress*, Prague Czech Republic.
22. Khalajmasoumi, M., **Sadeghi, B.** *, 2018 'Lithogeochemical anomaly prospectivity mapping of HREEs using multi-fractal modelling in Saghand Area, Central Iran', *International Association for Mathematical Geosciences (IAMG) Congress*, Prague Czech Republic.
23. **Sadeghi, B.** *, Carranza, E.J.M., Yilmaz, H., Ford, A., 2016 'Mapping of Au anomalies in drainage sediments by multifractal modelling', *35th International Geological Congress (IGC)*, Cape Town, South Africa; paper number 1286.

24. Khalajmasoumi, M., **Sadeghi, B.** *, Lotfi, M., 2016 ‘Discrimination and mapping of REEs litho-geochemical anomalies by multi-fractal modelling’, *35th International Geological Congress (IGC)*, Cape Town, South Africa; paper number 1287.
25. Rasi, R., Afzal, P., Yasrebi, A.B., Wethereld, A., **Sadeghi, B.** *, Mohammadi, B., 2016 ‘Application of fractal modelling for integration of data layers in GIS, Case study: Sharafabad-Hizehjan area (NW Iran)’, *35th International Geological Congress (IGC)*, Cape Town, South Africa- paper number 74.
26. **Sadeghi, B.** *, Carranza, E.J.M., Wang, H., Yilmaz, H., Ford, A., 2016 ‘Enhancement of Au Anomalies: Weight Bulk Leach Extractable Gold Data with Catchment Area’, *Goldschmidt2016*, Yokohama, Japan.
27. **Sadeghi, B.** *, Carranza E.J.M., 2015 ‘Improving geological logs of drill-cores by correlating with fractal models of drill-hole geochemical data’, *International Association for Mathematical Geosciences (IAMG) Congress*, Freiberg (Saxony), Germany (ISBN 978-3-00-050337-5).
28. Momeni, S., Afzal, P., **Sadeghi, B.**, 2014 ‘Separation of geochemical anomalies for prospecting of chromite mineralization using Concentration-Area fractal model in Khoy 1:100,000 sheet’, *5th Iranian Mining Engineering Congress*.
29. Rahmati, A., Afzal, P., Abrishamifar, S.A., **Sadeghi, B.**, Khodashenas Firouzabadi, N., 2013 ‘Comparison of the geological and grade distribution models in Sheytoor iron deposit, Bafq, Iran’, *23rd World Mining Congress*, Montreal, Canada, ISBN: 978-1-926872-15-5.
30. **Sadeghi, B.** *, Afzal, P., Moarefvand, P., KhodaShenas Firouzabadi, N., 2012 ‘Application of Concentration-Area fractal method for determination of Fe geochemical anomalies and the background in Zaghia area, Central Iran’, *34th International Geological Congress (IGC)*, Brisbane, Australia.
31. **Sadeghi, B.** *, Khalajmasoumi, M., Moarefvand, P., Afzal, P., 2012 ‘Extraction of iron alterations from ASTER SWIR and ETM⁺ data from Esfordi 1:100000 sheet (Central Iran)’, *International Mining Congress & Expo*, Tehran, Iran.
32. Katal, R., **Sadeghi, B.**, Adib, D, Memar KoochehBagh, A., 2012 ‘Geophysical sulphure zone exploration studies using IP & RS methods at Shadan area, Southern Khorasan province’, *International Mining Congress & Expo*, Tehran, Iran.
33. **Sadeghi, B.** *, Mehrabi, B., Eghbali, M.A., 2010 ‘General exploration of abandoned lead and zinc mine of RoknAbad in Ardabil province’, *International Mining Congress & Expo*, Tehran, Iran. (In Persian with English abstract).
34. Babaei, K., **Sadeghi, B.** *, 2010 ‘Discrimination of alteration zones from ASTER SWIR data in the lead and zinc abandoned area of RoknAbad in Ardabil province’, *International Mining Congress & Expo*, Tehran, Iran.

POSTER PRESENTATIONS:

1. **Sadeghi, B.** *, William D. Smith, 2025 ‘Unlocking automated mineralogy for accurate and consistent nomenclature of igneous cumulate rocks’, the 23rd *International Association for Mathematical Geosciences (IAMG) conference*, Zhuhai, China.
2. Teitler, Y., Lampinen, H., **Sadeghi, B.**, Iglesias-Martinez, M., Schoneveld, L., Verrall, M., Rashidi, B., 2025, Unravelling Rubidium Enrichment in Highly Fractionated Pegmatites: Insights from the High-Grade Mt Edon Deposit, Western Australia, SEG 2025 Conference, Brisbane, Australia.
3. **Sadeghi, B.** * and Cohen, D., 2023 ‘Discrimination of anthropogenic contamination and the effects of mineralization in soils from background patterns using multifractal modelling: Cyprus case study’, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-1585, <https://doi.org/10.5194/egusphere-egu23-1585>.

4. **Sadeghi, B. ***, Grunsky, E., 2022 ‘Compositional-Category-Based modeling: a new insight for geochemical anomaly classification’, *International Association for Mathematical Geosciences (IAMG) conference*, Nancy, France.
5. **Sadeghi, B. ***, Cohen, D.R., 2019c ‘Selecting the most robust geochemical classification model using the balance between the geostatistical precision and sensitivity’, *International Association for Mathematical Geosciences (IAMG) conference*, State College, Pennsylvania, USA (Conference Proceedings in Press).
6. **Sadeghi, B. ***, Cohen, D.R., Sadeghi, M., 2019d ‘Recognition and validation of complex geochemical anomalies related to VMS mineral deposits in Sweden’, *International Association for Mathematical Geosciences (IAMG) conference*, State College, Pennsylvania, USA (Conference Proceedings in Press).
7. Momeni, S., Shahrokhi, S.V., Afzal, P., Farhadinejad, T., **Sadeghi, B.**, 2013 ‘Separation of geochemical anomalies for prospecting of chromite mineralization using Concentration-Number fractal model in Khoy 1:100,000 sheet’, *First National Conference on Exploration Engineering of Underground Resources*, University of Shahrood, Iran.

RESEARCH PROJECTS

- **Sadeghi, B.**, 2014 ‘Geothermal energy resources in NW Iran, Eastern Azerbaijan’, provided for National Elite Foundation of Iran
- **Sadeghi, B.**, 2014 ‘The pre-feasibility study (PFS) project of the Sarkouh Cu-Mo deposit, Iran’; Project holder: National Iranian Copper Industries Co. (NICICO); Project consultant: Hampa Behineh Consultant Engineers Co.
- Afzal, P., Ziazarifi, A., **Sadeghi, B.**, 2013 ‘Reconnaissance of exploration targets for metallic mineralization by composition of airborne magnetometric, remote sensing and stream sediments geochemical data in Esfordi 1:100,000 sheet, Central Iran’; by supporting of Islamic Azad University, South Tehran branch.

BOOKS

- **Sadeghi, B. ***, 2024 ‘[Fractals and Multifractals in the Geosciences](#)’, *ELSEVIER* (May 17th, 2024; Paperback ISBN: 9780323908979 || eBook ISBN: 9780323908986), 302 pages (<https://doi.org/10.1016/C2020-0-03441-9>).
- Afzal, P., **Sadeghi, B.**, 2020 ‘Application of Fractal Methods in Geosciences and Mineral Exploration Engineering’, *IAU Press* (In Persian). *This book was developed based on my MSc thesis in addition to further case studies.*

BOOK CHAPTERS/ENTRIES

- Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F. 2023, Encyclopedia of Mathematical Geosciences, Springer, Cham, including:
 - **Sadeghi B. *** (2023) Variogram. In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1609–1613. https://doi.org/10.1007/978-3-030-26050-7_344-1
 - **Sadeghi, B. *** (2023): “Concentration-Area Plot” (Solo-authored). In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 169–175. https://doi.org/10.1007/978-3-030-26050-7_64-1

- **Sadeghi, B.** * (2023) Spectrum-Area Method. In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1398–1403. https://doi.org/10.1007/978-3-030-26050-7_310-1
- **Sadeghi, B.** * (2023): “Lotfi A. Zadeh biography” (Solo-authored). In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1643–1646. https://doi.org/10.1007/978-3-030-26050-7_385-1
- **Sadeghi B.** *, Ortiz J.M. (2023) Simulation. In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1322–1328. https://doi.org/10.1007/978-3-030-26050-7_292-1
- **Sadeghi, B.** *, Agterberg F. (2023) Singularity Analysis. In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1332–1339. https://doi.org/10.1007/978-3-030-26050-7_296-1
- **Sadeghi, B.** *, Grunsky E., Pawlowsky-Glahn V. (2023) Uncertainty Quantification. In: Daya Sagar B., Cheng Q., McKinley J., Agterberg F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 1583–1589. https://doi.org/10.1007/978-3-030-26050-7_334-1
- **Sadeghi, B.** *, Porwal, A., Pour, A.B., Rahimzadegan, M. (2023). Fuzzy Set Theory in Geosciences. In: Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham., 454–463. https://doi.org/10.1007/978-3-030-26050-7_10-1

SCHOLARLY RECOGNITION

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|---|------|
| ○ One of the most cited papers in Minerals over the last 36 months: Farhadi et al. (2022) ‘Combination of Machine Learning Algorithms with Concentration-Area Fractal Method for Soil Geochemical Anomaly Detection in Sediment-Hosted Irankuh Pb-Zn Deposit, Central Iran’, Minerals, vol. 12(6), 689. | 2025 |
| ○ Most popular paper of the journal from the last 3 years: Heidari et al. (2021) ‘Detection of mineralization stages using zonality and multifractal modeling based on geological and geochemical data in the Au-(Cu) intrusion-related Gouzal-Bolagh deposit, NW Iran’, Ore Geology Reviews, vol. 139, 104561. | 2024 |
| ○ Hot paper (7 th): Sadeghi, B. (2022) ‘Chatterjee Correlation Coefficient: A robust alternative for classic correlation methods in geochemical studies - (including “TripleCpy” Python package)’, Ore Geology Reviews, vol. 146, 104954. | 2022 |
| ○ Hot Paper (12 th): Zissimos et al. (2021) ‘Controls on soil geochemistry fractal characteristics in Lemesos (Limassol), Cyprus’, Journal of Geochemical Exploration, vol. 220. | 2021 |
| ○ Hot Paper (4 th): Afzal et al. (2013) ‘Separation of geochemical anomalies using factor analysis and concentration-number (C-N) fractal modelling based on stream sediments data in Esfordi 1:100,000 sheet, Central Iran’, Iranian Journal of Earth Sciences, vol. 5, pp. 100–110. | 2013 |
| ○ Hot Paper (8 th): Sadeghi et al., (2013) ‘Combination of geostatistical simulation and fractal modelling for mineral resource classification’, Journal of Geochemical Exploration, vol. 149, pp. 59–73. | 2015 |
| ○ Sadeghi et al. (2013) ‘Using ETM ⁺ and ASTER sensors to identify iron occurrences in the Esfordi 1:100000 mapping sheet of Central Iran’, Journal of African Earth Sciences, vol. 85, pp. 103–114. | 2013 |

- Hot Paper (5th, 14th, 15th, 16th) 2013
- Most cited articles since 2011 (25th), Journal of African Earth Sciences 2013
- Highly Cited Research certificate, Journal of African Earth Sciences 2016
- Sadeghi et al. (2012) ‘Application of fractal models to outline mineralized zones in the Zaghia iron ore deposit, Central Iran’, Journal of Geochemical Exploration, Special Issue "fractal/multifractal modelling of geochemical data", vol. 122, pp. 9–19.
- Hot Paper (14th) 2012
- The most cited articles since 2011 (19th), Journal of Geochemical Exploration 2012

LEADERSHIP, COMMUNITY ENGAGEMENT, & ACADEMIC SERVICE

- AIG official assessor for the RPGeo applications.
- Lead Session Convenor/Chair of “*Data-Driven Innovations for Mineral Exploration Decision-Making: Addressing Present and Future Challenges*” at IAMG, Zhuhai, China (2025)
- Scientific assessor of the oral and poster presentations at IAMG 2025, Zhuhai, China (2025)
- Scientific assessor of the oral and poster presentations at IAMG 2023, Trondheim, Norway (2025)
- Mentor at Australian Fulbright Alumni Association (AFAA) (2024-Present)
- The Association of Applied Geochemists ([AAG](#)) Councillor (2024 and 2025)
- The Association of Applied Geochemists (AAG) Strategy Planning Team (2017-2027)
- Lead Session Convenor of “Advancing Energy Transition-Associated Minerals Exploration: Data-Driven Ideas for Present and Future Challenges” at [GSA Connects 2024](#), Anaheim, California, USA (It was endorsed by I) JTPC Representative and II) Mineralogy, Geochemistry, Petrology, Volcanology (MGPV) Divisions).
- The IAMG Outreach Committee Member (2023-Present)
- The IAMG Liaison Committee Member (2023-Present)
- Associate Editor of *Ore Geology Reviews* (2020-Present)
- Associate Editor of *Geochemistry/Chemie der Erde* (2020-Present)
- Associate Editor of *Ore and Energy Resource Geology* (2020-Present)
- Editorial Board of *Applied Computing and Geosciences* (2024-Present)
- Editorial Board of *Earth Science Informatics* (2020-Present)
- Editorial Board of *Mining* (2020-Present)
- Editorial Board and Section Editor of *Minerals* (2021-Present)
- Editorial Board of Discover Minerals (2024-Present)
- Co-organizer of *MinProXT-2022 conference* (2022)
- Co-Guest Editor of [Geochemistry/Chemie der Erde](#) Special Issue “AI-driven exploration criteria identification and mineral prospectivity analysis” (2023-2024)
- Lead Guest Editor of *Geochemistry/Chemie der Erde* Special Issue “Critical Minerals: Current Perspectives” (2023-2024)
- Lead Guest Editor of *Minerals* Special Issue “Advanced Data Analytics in Earth and Environmental Sciences” (2022-2023)
- Lead Guest Editor of [Geochemistry/Chemie der Erde](#) Special Issue “Mineral exploration: a journey from fieldwork, to lab work, computational modeling and mineral processing” (2021)
- Solo Guest Editor of [Mining](#) Special Issue “Mathematical Geosciences in Exploration Geochemistry” (2021)

- Lead Session Convenor/Chair of “*Advances in computational analysis of geochemical survey data for geochemical anomaly detection*” at IAMG, Trondheim, Norway (2023)
- Lead Session Convenor/Chair of “*Data driven discovery in mineralogy and geochemistry: Data resources, analytics, and visualization*” at [EGU 2023](#), Vienna, Austria (2023)
- Lead Session Convenor/Chair on *Geochemical anomaly classification and modelling in mineral exploration* at [SGA 2022](#), New Zealand (2022)
- Sessions Convenor at *World Multi-disciplinary Earth Sciences Symposiums: WMESS 2016 and 2019* on Advanced mathematical-statistical modelling in exploration geochemistry (2016 & 2019)
- Geological Society of Australia Earth Sciences Student Symposium (GESSS 2019) (2019)
- Committee Member, Equity, Diversity and Inclusion (EDI) Subcommittee of GESSS 2019 (2019)
- Committee Member, Postgraduate Student Representative on the UNSW PANGAEA Research Centre Executive Committee (2019-2021)
- Mental and Physical Health First Aider at UNSW and USYD (2019-2021)
- Member of website and newsletter editorial board of Iranian Society of Mining Engineering (IRSME) (2011-2020)
- Student Volunteer Role, Goldschmidt 2016 (Japan) (2016)

THESIS, BOOK, JOURNAL, AND CONFERENCE REVIEWER ROLES

- Research proposal reviewer, invited by "Concurso de Proyectos Fondecyt de Iniciación en Investigación 2026", promoted by the National Agency for Research and Development under the Ministry of Science, Technology, Knowledge, and Innovation of Chile, July 2025.
- PhD Thesis proposal reviewer, invited by James Cook University, Australia, Jan 2025.
- *Book Proposal Reviewer*: “Mineral Exploration: Principles and Application to be authored by Swapan Kumar Halder”, May 2023, Elsevier.
- *Journal Reviewer*: Geoscience Data Journal, Artificial Intelligence in Geosciences, Scientific Reports, Applied Computing and Geosciences, Journal of Geophysical Research - Machine Learning and Computation, Earth Systems and Environment, Stats, Environmental Geochemistry and Health, Geology (GSA journal), Discover Minerals, Geological Journal, Fractal and Fractional, Catena, Mathematical Geosciences, Computers & Geosciences, Gondwana Research, Ore Geology Reviews, Natural Resources Research, Applied Geochemistry, Computational Geosciences, Renewable Energy, Frontiers in Marine Sciences, Environment, Development and Sustainability, Nonlinear Processes in Geophysics, Periodico di Mineralogia, Minerals, PLOSE ONE, Journal of Geochemical Exploration, Environmental Earth Sciences, Journal of African Earth Sciences, Earth Science Informatics, Geomatics, Journal of Mining Institute, Hacettepe Journal of Mathematics and Statistics, International Journal of Mining and Mineral Engineering, Journal of Geochemistry, Exploration, Environment, Analysis (GEEA), Advances in Space Research, SCIENCE CHINA Earth Sciences, Tethys, Arabian Journal of Geosciences, “CMSE” international congress in China and its peer-reviewed journal, Iranian Journal of Earth Sciences, Kharazmi Journal of Earth Sciences, Geopersia, Journal of Mining and Environment.
- *Conference Reviewer*: EGU 2023, International Conference on New Energy and Future System (NEFES 2016)

SOCIETY MEMBERSHIP

- Australian Fulbright Alumni Association (AFAA).

- The American Geophysical Union (AGU).
- The Geological Society of America (GSA).
- The Australasian Institute of Mining and Metallurgy (AusIMM): **Fellow**.
- The Australian Institute of Geosciences (AIG): **Fellow** and **RPGeo**.
- European Geosciences Union (EGU).
- Society for Geology Applied to Mineral Deposits (SGA).
- Geological Society of Australia (GSA).
- International Association for Mathematical Geosciences (IAMG): **Lifetime member**.
- The Association of Applied Geochemists (AAG): **Fellow**.
- Fulbright Australia (<https://fulbrighternetwork.com>).
- Austmine.
- Engineers Australia (EA).
- Iranian Society of Mining Engineering (IRSME): **Honorary Fellow**.
- Mental Health First Aid Australia.

PYTHON PACKAGES DEVELOPMENT

pyClusterWise: developed to select the most reliable clustering methods and evaluation metrics per data/scenario considering their uncertainties.

IRWPy: developed to apply my IRW interpolation method.

TripleCpy: newly developed Chatterjee Correlation Coefficient provider.

OUTREACH SERVICES

- The IAMG Outreach Committee Member (2023-Present)
- Author of educational materials targeting children (ages 6 to 10) to impart scientific knowledge through songs; Currently in the process of seeking publication through various U.S. publishers (2023)
- Undergraduate student mentor, “*Volcanic, sedimentary, and metasomatic rocks, explaining their origins and distinctions*”, University of Kurdistan (2006)
- High-school student mentor, “*What is geology and mineral exploration and why it matters?*”, University of Kurdistan (2005)

IN THE MEDIA

- **Carnegie Science**, Washington DC, USA
- **CSIRO (ARRC)**, Australia
- IAMG 2023 Andrei Borisovich Vistelius Research Award
 - **LinkedIn**
 - **Twitter**
 - **Facebook**
 - **Instagram**
 - **IAMG**
 - **Wikipedia**
- MineTED:
 - **Original webpage**
 - **YouTube**
- Fulbright Scholars:
 - **Fulbright Australia**
 - **UNSW**
 - **USYD**

- [EarthByte Group](#)
- IAMG Award Winner's Speech 2021: [IAMG](#)
- IAMG Award Winners: [IAMG](#)
- IAMG [Founders Scholarship](#)
- IAMG Newsletters:
 - No. [99](#)
 - No. [100](#)
- [PANGEA Research Centre](#) (UNSW)-Now "Earth and Sustainability Science Research Centre: ESSRC"
- EarthByte Group, USYD:
 - [Twitter](#)
 - [Facebook](#)
 - [Website: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10](#)
 - [University of Sydney Marine Studies Institute](#)